



SCENARIO-BASED ELEARNING

Coral Reefs on the Brink

A Field Scientist's First Decision

You don't learn reef conservation from a textbook. You learn it by getting to know the reef, reading its warning signs, weighing a real call when you cannot save everything, and finding your place in the work, which is what this course puts you through.

WHAT IT IS

A four-part, scenario-based course that introduces reef conservation by letting the learner do the work, not read about it: understanding the living reef, diagnosing heat stress, making a real management call under pressure, and finding where they fit in the field. A flight simulator for the job, not a textbook about it.

FORMAT / TOOL

Articulate Rise 360 with an embedded, variable-driven Storyline 360 simulation and several custom HTML/CSS/JS code blocks. Four modules.

STATUS

Fully built and playable. Completed and confirmed by test users.

WHO IT IS FOR

Aspiring marine scientists from advanced high school to first-year university, and anyone weighing the field who wants to find out whether reef conservation work is for them.

LENGTH / LANGUAGE

Self-paced. Roughly 50–70 min of core seat time; about 1.5–2 h for a thorough run. English.

The Starting Point and the Goal

The course sets out to do one thing: introduce someone to reef conservation by putting them inside the work rather than in front of a reading list. It runs the full arc, what a living reef is and why it is worth protecting, how to read the warning signs of heat stress, what it takes to make a real call when resources fall short, and where a newcomer might fit in the field. That path, from the science to a career direction, is the design, not an add-on, and it is what makes the course useful to universities that want to draw new people toward reef science.

Its dramatic high point is a Code Red marine heatwave: a three-zone reef site, ten days above the bleaching threshold, limited resources, and three stakeholders who each want something different. Here the goal sharpens to the hardest part of the job, deciding what to save first when you cannot save it all, and defending that call with evidence.

The science underneath is not set dressing; at the reef it is exact mechanism that has to be right. I am trained both in molecular biology (PhD) and in tropical marine ecology, where I focused on coral reefs at the University of Bremen and the Leibniz Centre for Tropical Marine Research (ZMT), so I could build the scientific backbone, the coral-algae symbiosis, the bleaching cascade, the Degree Heating Weeks logic, to be correct and judge whether it holds rather than letting it merely sound plausible. Where the course simplifies for clarity, it says so on the screen.

What I Built

A single continuous arc across four modules, moving the learner from wonder to analysis to a decision under pressure, and finally to their own next step. The emotional shape is deliberate: Awe, Analysis, Action, Personalization.

1 **Awe.** The Living City (Module 1).

The reef is established as a living system worth fighting for: the coral-algae partnership that powers it, the three reef types and what each does for the coast, and an interactive “Living Breakwater” simulation where restoring the grazer-coral balance visibly brings wave protection back. Curiosity first, stakes second.

2 **Analysis.** The Silent Alarm (Module 2).

The diagnostic toolkit. The learner sequences the bleaching cascade, learns that a pale coral is starving but still alive, the recovery window the whole triage logic later hinges on, and reads thermal-stress data in a custom Degree Heating Weeks widget that sorts sites into Watch, Warning, and Alert. Concern, turned into diagnostic confidence.

3 **Action.** The Decision Lab (Module 3, the centerpiece).

A continuous, variable-driven Storyline simulation. As a newly appointed reef manager in Code Red, the learner forms a gut-check hypothesis, then confronts it with live data to diagnose the highest-risk zone; works through three stakeholder dossiers (field operations, the tourism board, the regional director) to find the dependencies between them; sequences a response, deploy, escalate, draft the closure, where each step unlocks the next; and makes a hard secondary call for a zone that cannot get full coverage. Seventy-two hours later, a consequence report shows what the decision cost and saved, including the case where an unwatched zone collapses unseen. Wrong choices are never dead ends: the simulation explains the principle and lets the learner try again.

4 **Personalization.** Your Dive into the Field (Module 4).

The experience is turned into a direction. The learner meets six real conservation roles and what each one actually produces, names the career barriers most relevant to them (funding, geography, time, knowledge gaps) and receives adaptive strategies for each, and builds a personalized 30-day action plan with concrete first steps and real resources, from citizen-science programs to NOAA Coral Reef Watch. The course ends not on a score but on a plan.

The Design Approach

The course rests on one conviction: judgment is practiced, not told. Three choices follow from it.

- **A real call, not a quiz.** The simulation’s options are operationally plausible, and the feedback is intrinsic and delayed. Instead of “correct” or “incorrect,” the learner watches the consequences unfold; a wrong sequence triggers a stakeholder refusal in character, not a buzzer. The hardest lesson, that some assets cannot be saved and that prevention rarely gets credit, is built into the outcome rather than announced.
- **Designed for the emotional state, not just the content.** Reef science carries real eco-anxiety. The course is deliberately hopeful and agency-focused, placing “Hope Anchors” such as evidence of localized recovery so the crisis reads as difficult but solvable. The target is a learner who feels able to act, not one who feels paralyzed.
- **Rigor that is honest about its own limits.** Where the course simplifies, a three-step risk scale in place of the full one, a single hallmark service per reef type, it says so on screen. Being open about what is simplified is itself a piece of scientific integrity, and it is the part most courses quietly skip.

The Learning Principles Behind It

- **Mastery learning:** progression is gated on demonstrated competence, not a rank. Everyone who finishes has actually performed the target reasoning.
- **Intrinsic, delayed feedback:** the “72 Hours Later” report is the teacher. The learner lives with the call rather than being corrected out of it.
- **Decision-making under uncertainty:** a gut-check hypothesis is committed before the data arrives, then tested against it, the move that separates reasoning from guessing.
- **Triage ethics:** the simulation refuses the comfortable idea that everything is savable, and names the “Prevention Trap,” the reason careful, preventive calls so often go unrewarded.
- **Metacognition:** confidence sliders before and after the hard parts let the learner see their own growth in self-efficacy, not just a score.
- **Accessibility by default:** WCAG-aligned, keyboard-navigable interactions, with the risk scale encoded in color, pattern, and icon together, so meaning never rests on color alone.
- **Authentic artifacts:** the learner leaves with five usable takeaways, a Reef 101 one-pager, a bleaching quick guide, a field decision record, a 30-day action plan, and a summary dashboard, rather than a completion mark.

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